

MACHINE LEARNING FUNDAMENTALS SERIES

Neural Networks Pt. 3: Backpropagation Details Pt. 1

Simultaneous Optimization of Weights w_3, w_4 and Bias b_3 via Chain Rule & Gradient Descent

Source: StatQuest with Josh Starmer — Backpropagation Details Pt. 1: Optimizing 3 parameters simultaneously.

1. Introduction and Problem Formulation

In the previous module, we demonstrated how **Backpropagation** optimizes a single final bias term b_3 while keeping all other parameters fixed. In this module, we advance to optimizing **three parameters simultaneously**: two connection weights w_3, w_4 and the output bias b_3 .

Pedagogical Scope & Mathematical Notation

The primary objective of this module is to formalize the **Chain Rule** and **Gradient Descent** for multi-parameter optimization using rigorous mathematical notation:

- **Mathematical Notation:** We replace full textual labels with formal mathematical variables ($Input_i, x_{1,i}, x_{2,i}, a_{1,i}, a_{2,i}, w_3, w_4, b_3, \hat{y}_i$).
- **Pre-optimized Parameters:** $w_1 = 3.34, b_1 = -1.43, w_2 = -3.53$, and $b_2 = 0.57$ are fixed at their optimal values.
- **Parameters to Optimize:** w_3, w_4 , and b_3 are treated as unknown variables to be optimized simultaneously.

2. Neural Network Topology and Variable Definitions

Consider a training dataset with $n = 3$ drug dosage samples $Input_i \in \{0.0, 0.5, 1.0\}$ and observed effectiveness targets $y_i \in \{0.0, 1.0, 0.0\}$.

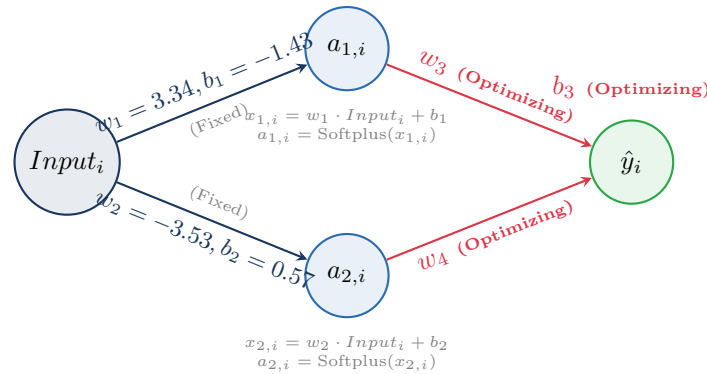


Figure 1: Minimalist Neural Network Architecture highlighting the 3 parameters (w_3, w_4, b_3) under simultaneous backpropagation optimization.

2.1. Formal Indexing and Activation Formulas

For each sample $i \in \{1, \dots, n\}$:

1. Top Node Input & Activation:

$$x_{1,i} = w_1 \cdot Input_i + b_1 = 3.34 \cdot Input_i - 1.43, \quad a_{1,i} = \text{Softplus}(x_{1,i}) = \ln(1 + e^{x_{1,i}}) \quad (1)$$

2. Bottom Node Input & Activation:

$$x_{2,i} = w_2 \cdot Input_i + b_2 = -3.53 \cdot Input_i + 0.57, \quad a_{2,i} = \text{Softplus}(x_{2,i}) = \ln(1 + e^{x_{2,i}}) \quad (2)$$

3. Output Layer Prediction:

$$\hat{y}_i = w_3 \cdot a_{1,i} + w_4 \cdot a_{2,i} + b_3 \quad (3)$$

3. Loss Function and Chain Rule Derivatives

To evaluate model fit quality across the dataset, we calculate the **Sum of Squared Residuals (SSR)**:

$$\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

Since $\hat{y}_i$ directly links w_3, w_4 , and b_3 to SSR, we apply the **Chain Rule** to derive partial derivatives with respect to each parameter.

3.1. Partial Derivative with Respect to b_3

By Chain Rule:

$$\frac{\partial \text{SSR}}{\partial b_3} = \sum_{i=1}^n \frac{\partial \text{SSR}}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial b_3} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot 1 = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \quad (5)$$

3.2. Partial Derivative with Respect to w_3

By Chain Rule:

$$\frac{\partial \text{SSR}}{\partial w_3} = \sum_{i=1}^n \frac{\partial \text{SSR}}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w_3} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot a_{1,i} \quad (6)$$

3.3. Partial Derivative with Respect to w_4

Similarly, since $\frac{\partial \hat{y}_i}{\partial w_4} = a_{2,i}$:

$$\frac{\partial \text{SSR}}{\partial w_4} = \sum_{i=1}^n \frac{\partial \text{SSR}}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w_4} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot a_{2,i} \quad (7)$$

Key Insight on Derivative Structure

Notice that all three parameter gradients share the exact same core error term $\frac{\partial \text{SSR}}{\partial \hat{y}_i} = -2(y_i - \hat{y}_i)$. The only difference is the scaling factor $\frac{\partial \hat{y}_i}{\partial \theta}$: 1 for bias b_3 , $a_{1,i}$ for weight w_3 , and $a_{2,i}$ for weight w_4 .

4. Parameter Initialization and Step-by-Step Optimization

Following standard practice, we initialize parameters as follows:

- $w_3^{(0)} = 0.36$ (sampled from a standard normal distribution $\mathcal{N}(0, 1)$).
- $w_4^{(0)} = 0.63$ (sampled from $\mathcal{N}(0, 1)$).
- $b_3^{(0)} = 0.0$ (bias initialized at zero).

4.1. Forward Pass Calculations at Step 0

Evaluating activation values for each dosage Input_i :

Table 1: Forward Pass Activation Values and Initial Predictions ($w_3 = 0.36, w_4 = 0.63, b_3 = 0.0$)

Sample i	Input_i	$a_{1,i}$	$a_{2,i}$	$\hat{y}_i$	y_i	Residual ($y_i - \hat{y}_i$)
1	0.0	0.222	1.026	0.72	0.0	-0.72
2	0.5	0.083	0.347	0.25	1.0	+0.75
3	1.0	0.134	0.051	0.08	0.0	-0.08

Initial Loss: $\text{SSR}^{(0)} = (-0.72)^2 + (0.75)^2 + (-0.08)^2 \approx \mathbf{1.08}$.

4.2. Gradient Evaluation and Step Size Computation

Plugging values into our partial derivative formulas:

$$\frac{\partial \text{SSR}}{\partial w_3} = -2 \left[(-0.72)(0.222) + (0.75)(0.083) + (-0.08)(0.134) \right] = \mathbf{0.22} \quad (8)$$

$$\frac{\partial \text{SSR}}{\partial w_4} = -2 \left[(-0.72)(1.026) + (0.75)(0.347) + (-0.08)(0.051) \right] = \mathbf{0.96} \quad (9)$$

$$\frac{\partial \text{SSR}}{\partial b_3} = -2 \left[(-0.72) + (0.75) + (-0.08) \right] = \mathbf{0.10} \quad (10)$$

4.3. Simultaneous Parameter Updates

Using learning rate $\eta = 0.1$, all 3 parameters are updated simultaneously in parallel:

$$w_3^{(1)} = w_3^{(0)} - \eta \cdot \frac{\partial \text{SSR}}{\partial w_3} = 0.36 - (0.1 \times 0.22) = \mathbf{0.338} \quad (11)$$

$$w_4^{(1)} = w_4^{(0)} - \eta \cdot \frac{\partial \text{SSR}}{\partial w_4} = 0.63 - (0.1 \times 0.96) = \mathbf{0.534} \quad (12)$$

$$b_3^{(1)} = b_3^{(0)} - \eta \cdot \frac{\partial \text{SSR}}{\partial b_3} = 0.00 - (0.1 \times 0.10) = \mathbf{-0.010} \quad (13)$$

4.4. Iterative Gradient Descent Convergence

This multi-parameter update process repeats iteratively. After 175 steps in Gradient Descent, the parameters converge to their optimal values:

$$w_3^* \rightarrow -1.30, \quad w_4^* \rightarrow +2.28, \quad b_3^* \rightarrow -0.58 \implies \text{SSR minimized} \quad (14)$$

5. Summary of Multi-Parameter Backpropagation

Table 2: Comparative Overview of Backpropagation Derivatives across Network Layers

Parameter	Chain Rule Expression	Scaling Factor ($\frac{\partial \hat{y}_i}{\partial \theta}$)
Output Bias b_3	$-2 \sum (y_i - \hat{y}_i) \cdot 1$	1 (Identity)
Output Weight w_3	$-2 \sum (y_i - \hat{y}_i) \cdot a_{1,i}$	$a_{1,i}$ (Top Activation)
Output Weight w_4	$-2 \sum (y_i - \hat{y}_i) \cdot a_{2,i}$	$a_{2,i}$ (Bottom Activation)